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Guerra

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(54) DROP-IN SYSTEM TO ENHANCE THE RESETTING AND OPERATIONS OF SEMIAUTOMATIC FIREARMS

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F41A 17/48 (2006.01)

(52) **U.S. Cl.**
CPC *F41A 19/15* (2013.01); *F41A 17/48* (2013.01)

(58) **Field of Classification Search**
CPC F41A 19/15; F41A 19/42; F41A 19/44; F41A 17/48; F41A 11/00
See application file for complete search history.

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(57) **ABSTRACT**

A drop-in system for a semiautomatic firearm trigger includes a trigger-locking block, a hump block, a trigger-locking block transfer bar and an axis retainer plate that operate together to enhance the operations of a semiautomatic firearm trigger. The trigger-locking block is pivoted on a firearm safety pin to lock the trigger when the carrier bolt travels rearward, the hump block is pivotally disposed within a trigger cradle on a trigger pin to receive a force of a hammer to rotate rearward to rotate the trigger assembly backward, the trigger-locking block transfer bar slides along carrier bolt guide and rotate the trigger-locking block forward to releasee the trigger, and the axis retainer plate holds the trigger-locking block rotatably in place and limits the rotation of the trigger-locking block.

6 Claims, 8 Drawing Sheets

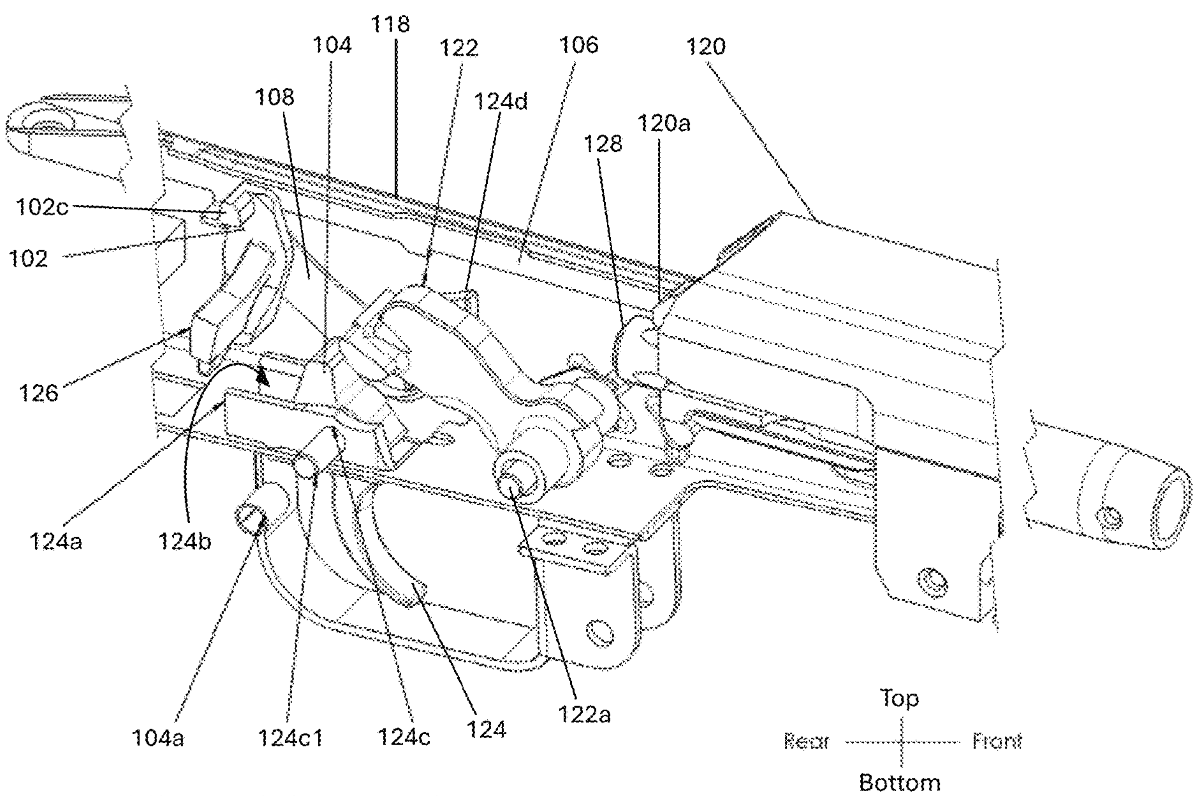


FIG. 1A
(PRIOR ART)

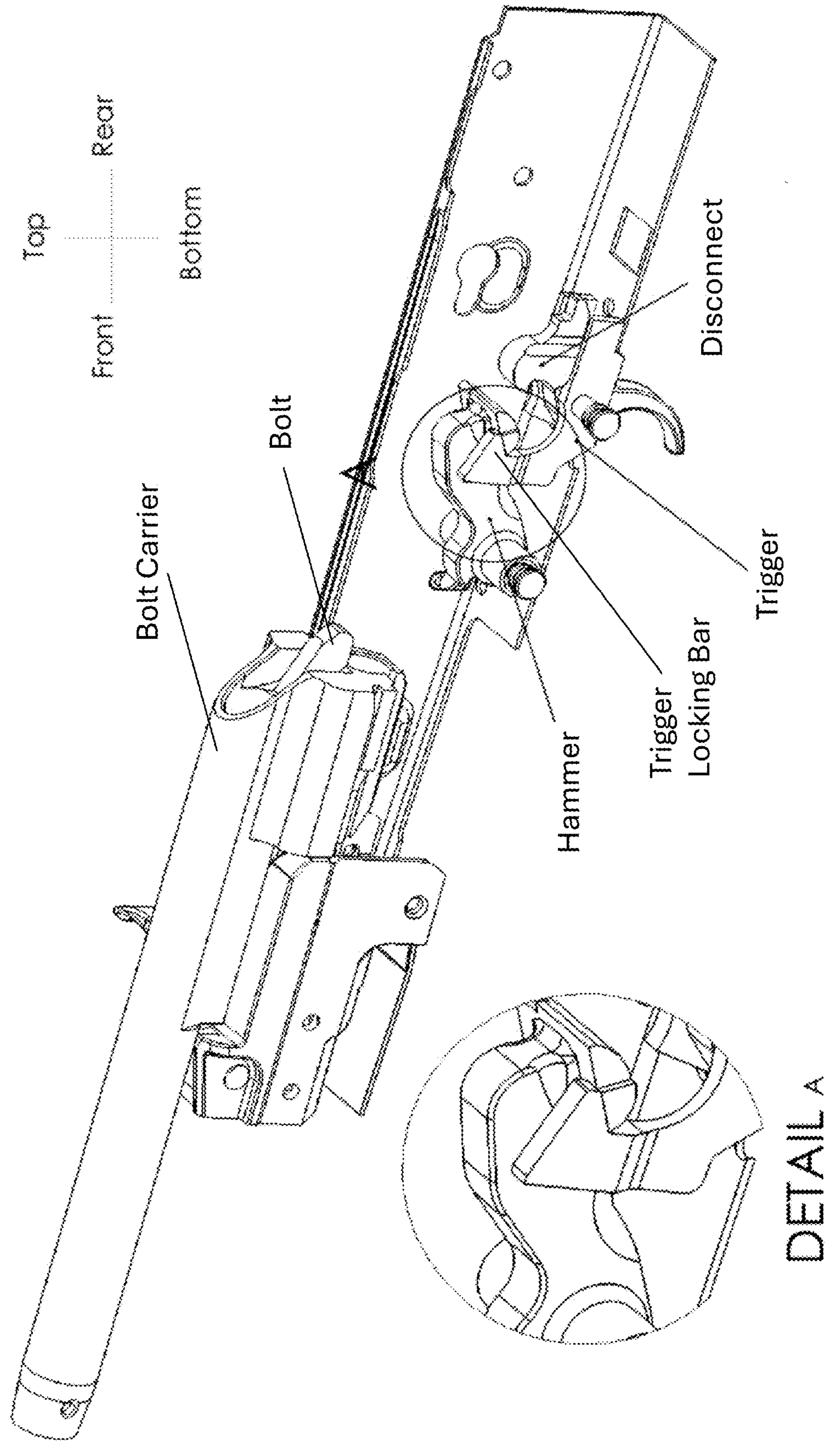
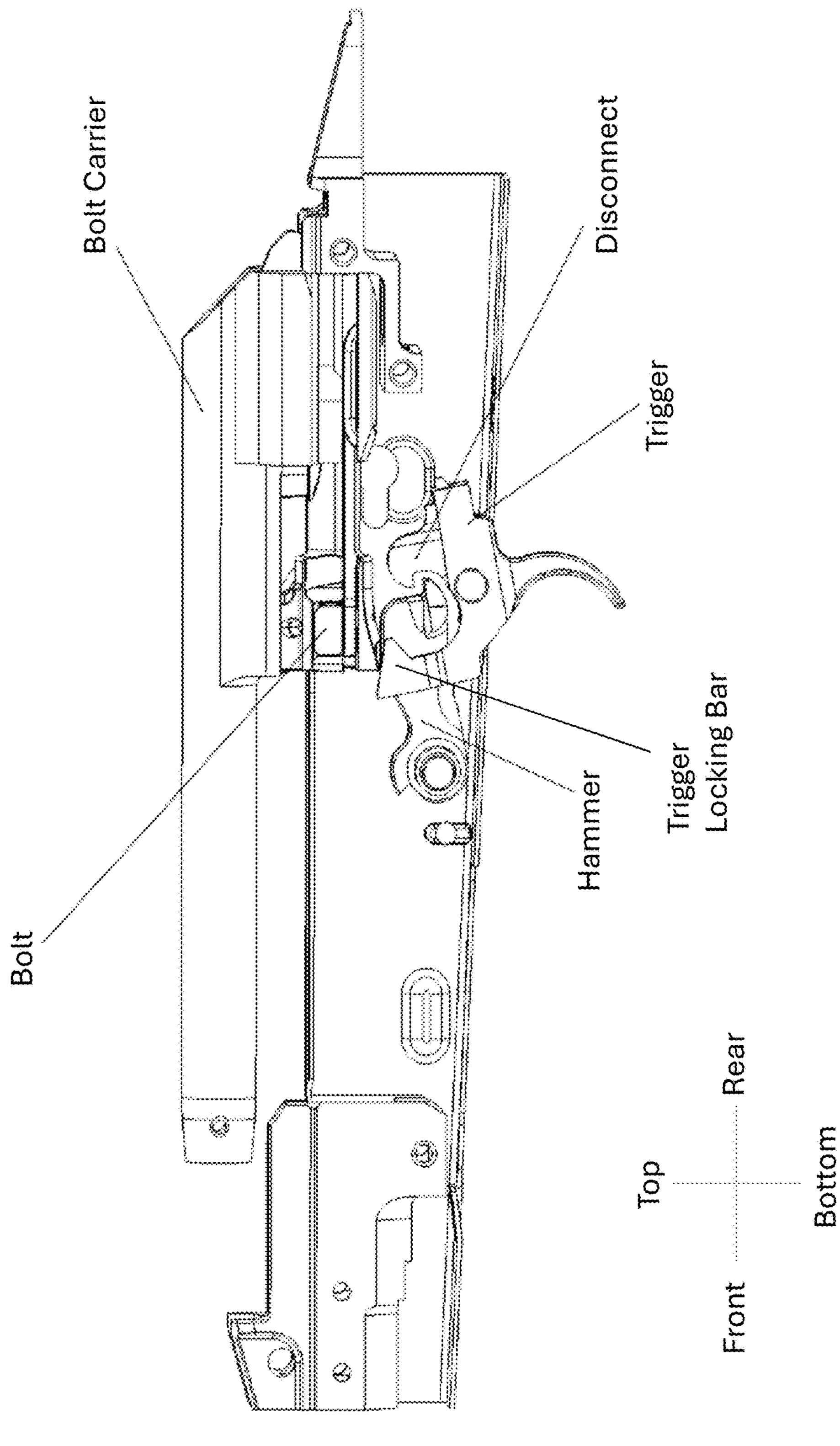


FIG. 1B
(PRIOR ART)



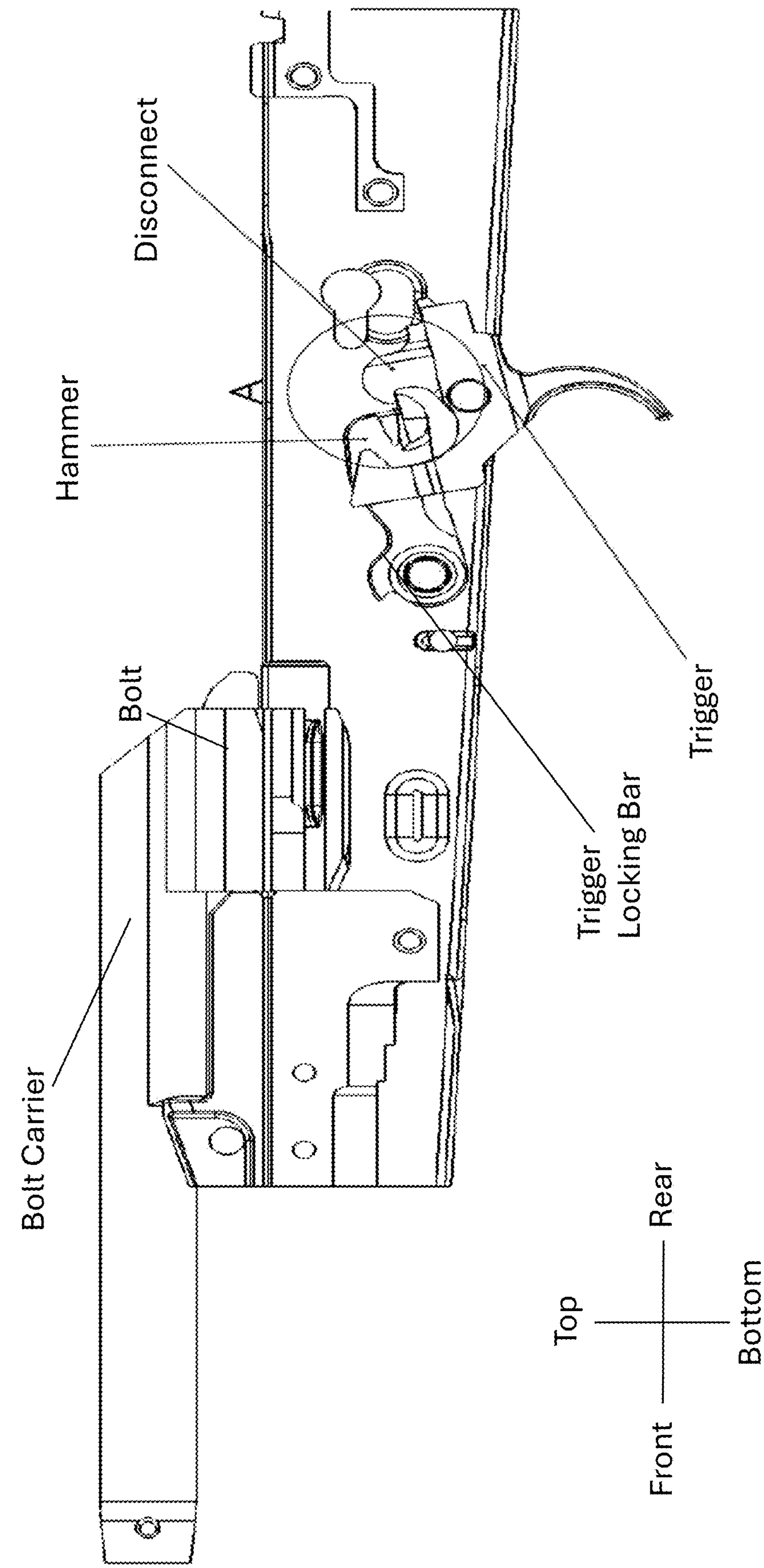
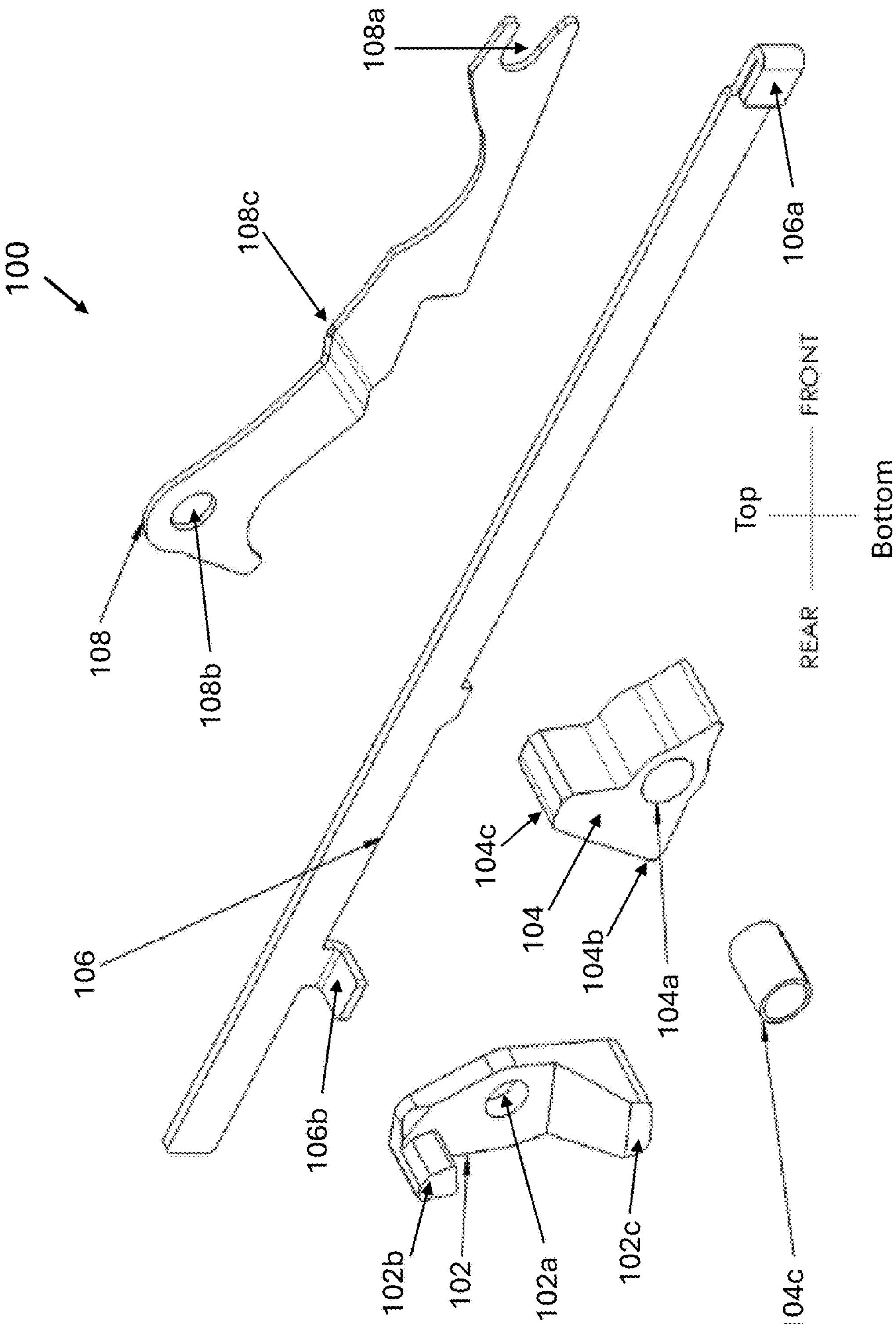


FIG. 1C
(PRIOR ART)

FIG. 2


100
108a
108c
108
108b
106a
106
106b
104c
104
104b
104a
102b
102
102a
102c
104c
Top
Bottom
FRONT
REAR

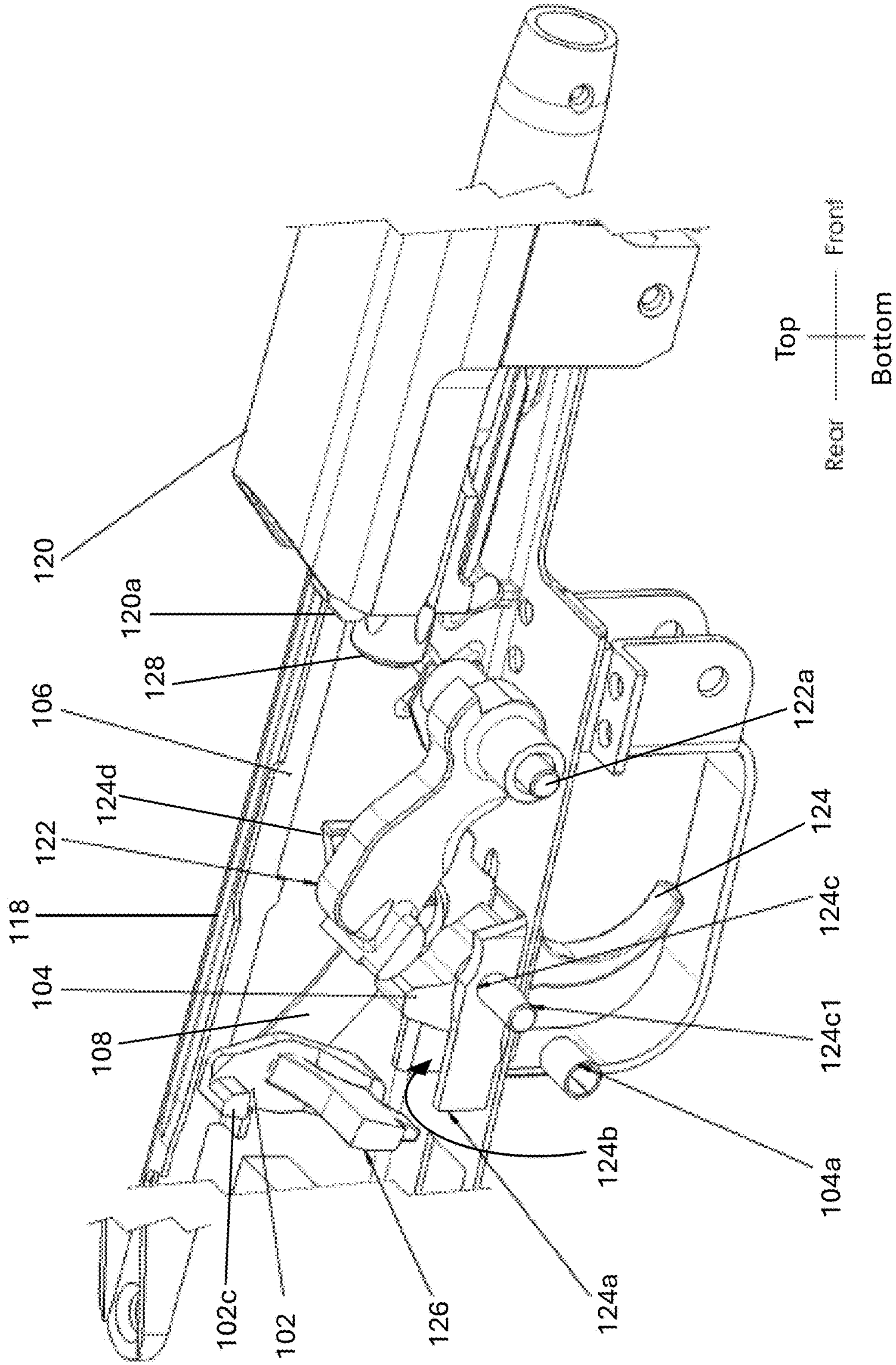
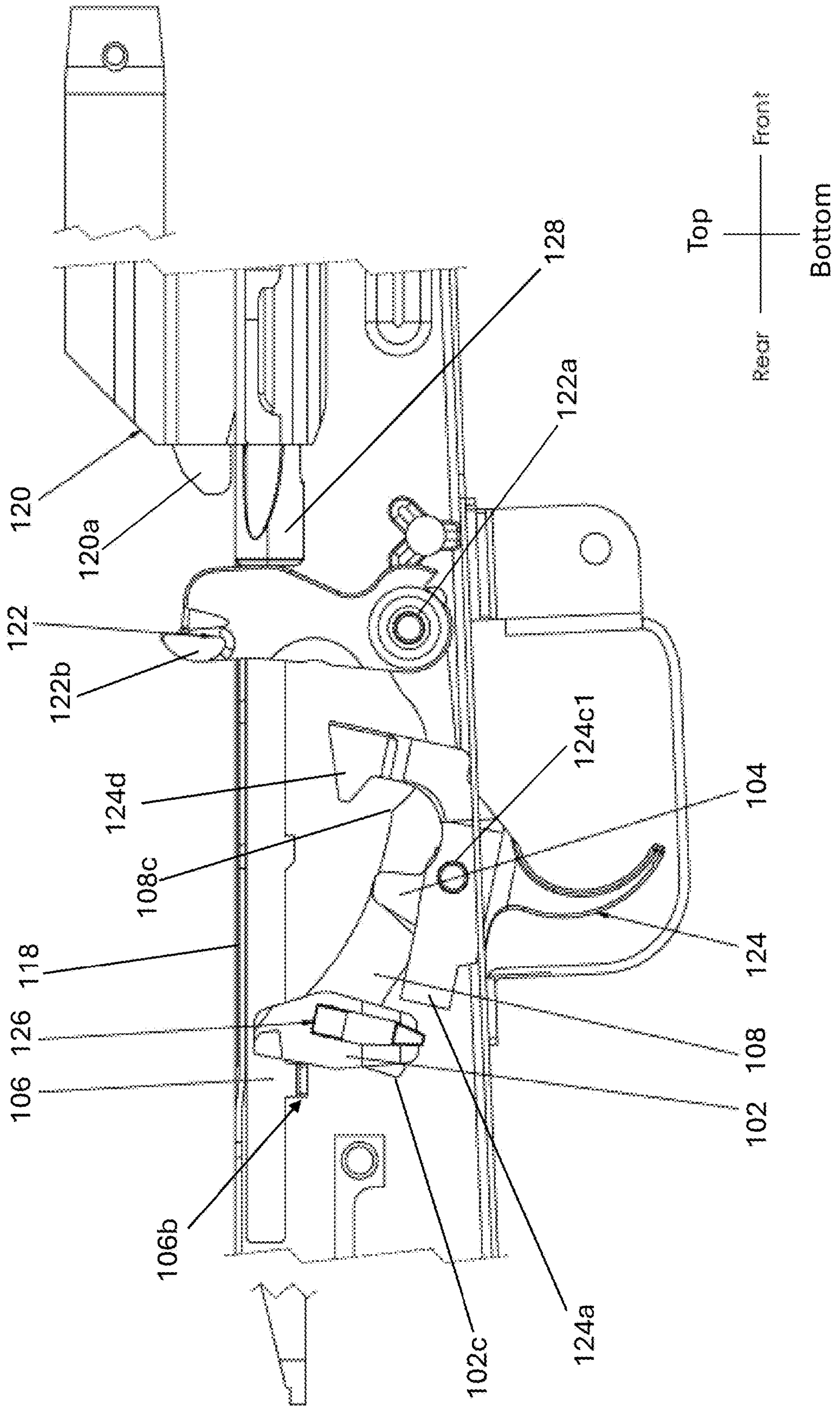

120
120a
128
106
124d
122
118
104
108
102c
102
126
124a
124b
104a
124c1
124c
124
122a
Top
Bottom
Rear
Front

FIG. 3



120
120a
128
122a
122
122b
124d
124c1
104
108c
118
124
126
108
106
102
106b
102c
124a
Top
Bottom
Front
Rear

FIG. 4

FIG. 5

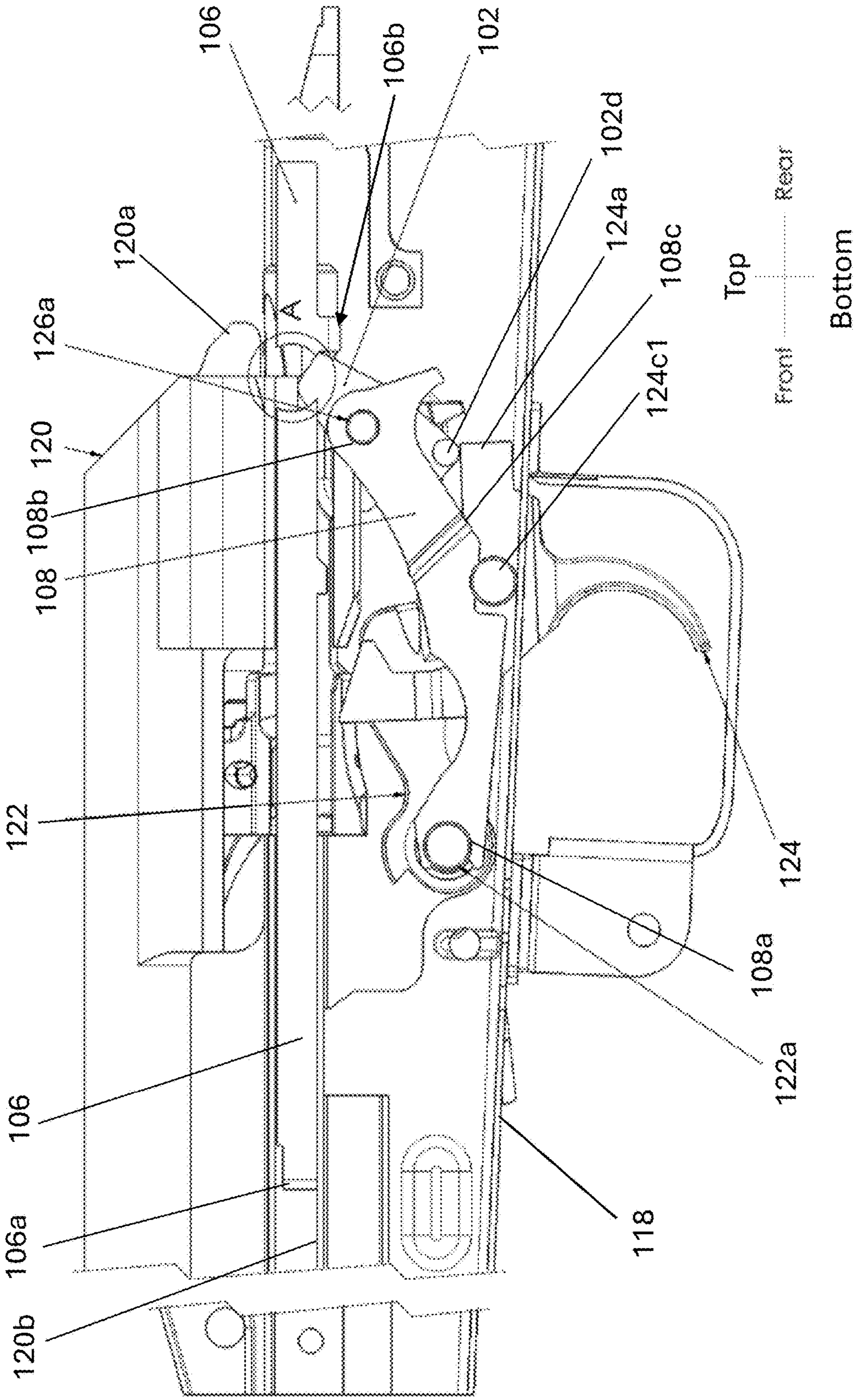
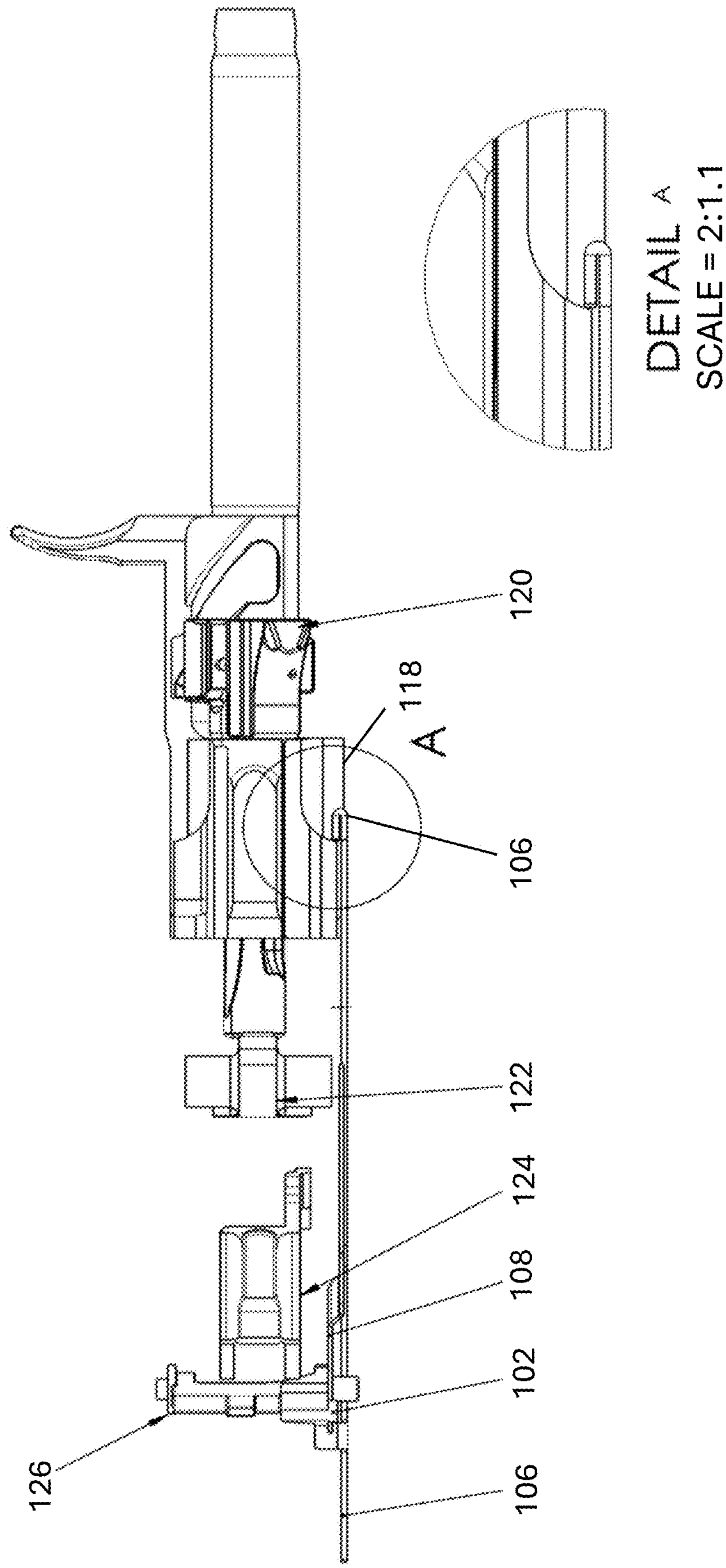

106
106b
102
102d
124a
108c
124c1
120a
126a
120
108b
108
A
Top
Bottom
Front
Rear
122
124
108a
122a
106
118
106a
120b

FIG. 6



DROP-IN SYSTEM TO ENHANCE THE RESETTING AND OPERATIONS OF SEMIAUTOMATIC FIREARMS

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

Not applicable.

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BACKGROUND OF THE INVENTIVE CONCEPT

1. Field of the Invention

The present inventive concept relates to a drop-in system for semiautomatic firearms, and more particularly, to a drop-in system to enhance the resetting and operations of a trigger for semiautomatic firearms.

2. Description of the Related Art

A standard semiautomatic firearm generally includes a trigger, a trigger-locking bar, a hammer, a bolt carrier with a bolt, and a disconnect. FIGS. **1A-1C** illustrate basic operations of the standard semiautomatic firearm.

Referring to FIG. **1A**, the trigger includes a trigger-locking bar, which holds a hammer in a cocked position until a trigger is pulled. As illustrated in FIG. **1A**, the trigger is at a reset position where the trigger is holding the hammer back in a cocked position ready to be released in a forward direction to fire the semiautomatic firearm. If the well-known firearm safety switch (not illustrated) is in an active position the trigger will be prevented from being pulled to release the hammer, and if the firearm safety is in a non-active position the trigger will be enabled to be pulled.

FIG. **1B** illustrates the stage where the firing cycle of the semiautomatic firearm is at a halfway point with the trigger in a pulled position. At this stage the semiautomatic firearm is in a state where the hammer is being released from the trigger locking bar to fire the firearm.

FIG. **1C** illustrates a state in which the trigger is still in the depressed position and the disconnect catches the hammer on the way back after the semiautomatic firearm has been fired. The main job of the disconnect is to hold down the hammer to prevent the hammer from following the bolt carrier forward when the bolt carrier is moving back to its forward position after driving the hammer back. As a result of a spring action the bolt carrier is forced back after the semiautomatic is fired. The disconnect is provided to catch the hammer when the hammer is pushed back by the bolt carrier since the trigger may still be in the pulled. This is important since if the hammer follows that bolt carrier forward while the bolt carrier is traveling forward and the trigger is still in the pulled position the hammer, although traveling forward, will not have enough kinetic energy to set off the primer when sticking a firing pin (not illustrated), which sets off the cartridge in the barrel. When the pressure being applied to the trigger is released the disconnect will also follow the travel movement of the trigger, and the trigger will catch the hammer to "reset" the firearm while the disconnect releases its hold on the hammer. This action of the trigger catching the hammer results in the semiautomatic firearm returning back to the position as illustrated in FIG. **1A**.

SUMMARY OF THE PRESENT INVENTIVE CONCEPT

The present general inventive concept. More particularly, but not exclusively, the present inventive concept a drop-in system for semiautomatic firearms, and more particularly, to a drop-in system for semiautomatic firearms to enhance the resetting and operations of a trigger for semiautomatic firearms.

Additional features and utilities of the present general inventive concept will be set forth in part in the description which follows and, in part, will be obvious from the description, or may be learned by practice of the general inventive concept.

The foregoing and/or other features and utilities of the present general inventive concept may be achieved by providing a drop-in system for a semiautomatic firearm that includes a trigger assembly pivotal about a trigger axis pin, a bolt carrier that slides back and forth along a receiver of the semiautomatic firearm, a spring-loaded hammer that rotates about a hammer axis pin, a firearm safety including a corresponding safety axis pin extending therefrom, and a bolt carrier guide that extends along an interior of the receiver to guide the bolt carrier back and forth, the drop-in system comprising: an elongated axis retainer plate having one end configured to be fixed to the hammer axis pin and a second end configured to be fixed to the safety axis pin; a trigger-locking block configured to pivot about the firearm safety axis pin, the trigger-locking block including a limit protrusion extending from a lower end thereof to contact the elongated axis retainer plate to limit rearward pivotal movement of the trigger-locking block; a hump block configured to pivot about the trigger pin independently of pivotal movement of the trigger assembly; and a trigger-locking block transfer bar configured to slide along the bolt carrier guide and including a limit protrusion extending from a rear bottom portion thereof.

In an example embodiment, when the bolt carrier travels in a rearward direction along the receiver due to firing of the semiautomatic firearm: the bolt carrier collides with the hammer; the hammer rotates rearward and collides with the hump block; the hump block pivots rearward and collides with the trigger assembly to rotate the trigger assembly rearward; and the bolt carrier collides with the trigger-locking block to rotate the trigger-locking block rearward to lock a rear end of the trigger assembly.

In another example embodiment, after the bolt carrier completes the travel rearward the bolt carrier springs in a forward direction to collide with the limit protrusion of the trigger-locking block transfer bar to slide the trigger-locking bar in a forward direction such that the limit protrusion collides with a top portion of the trigger-locking block to pivot the trigger-locking block in a forward direction to release the rear end of the trigger assembly from the locked position.

In another example embodiment, the elongated axis retainer plate comprises: a notch formed at a first end thereof to connect with the hammer axis pin; a hole extending through a second opposite end thereof to receive the safety axis pin therethrough; and a bend formed at a middle portion thereof to extend inward and apply a friction to a side of the trigger-locking block and to stop the limit protrusion to limit the rearward rotation of the trigger-locking block. In still another example embodiment, the elongated axis retainer plate is disposed in the receiver by the notch formed at the first end thereof being fixed over the hammer axis pin and the safety axis pin being slid through a first hole extending through a first side of the receiver, through a hole extending through the trigger-locking block, through the hole extending through the second end of the elongated axis retainer plate, and an end of the safety axis pin being snapped into a second hole extending through a second side of the receiver.

The foregoing and/or other features and utilities of the present general inventive concept may also be achieved by providing a drop-in system for a semiautomatic firearm configured to be placed within a receiver of the semiautomatic firearm, comprising: an axis retainer plate configured to be disposed between a hammer axis pin and safety axis pin; a hump block configured to pivot about a trigger axis pin of a trigger to pivot the trigger backward; a trigger-locking block configured to pivot about the safety axis pin in a first direction until making contact with the elongated axis retainer plate and locking the trigger from being pulled; and a trigger-locking block transfer bar configured to slide along an inner wall of the receiver and including a limit protrusion extending from a rear bottom portion thereof to pivot the trigger-locking block in a second direction to unlock the trigger.

Additional features and utilities of the present general inventive concept will be set forth in part in the description which follows and, in part, will be obvious from the description, or may be learned by practice of the general inventive concept.

BRIEF DESCRIPTION OF THE DRAWINGS

These and/or other features and utilities of the present inventive concept will become apparent and more readily appreciated from the following description of the embodiments, taken in conjunction with the accompanying drawings of which:

FIGS. **1A-1C** illustrate various steps of a semiautomatic firearm according to conventional operations.

FIG. **2** illustrates an exploded perspective view of a drop-in system for a firearms trigger, according to an example embodiment of the present inventive concept.

FIG. **3** illustrates an operational perspective view of the drop-in system for a firearms trigger, according to the example embodiment illustrated in FIG. **2**.

FIG. **4** illustrates an operational side view of the drop-in system for a firearms trigger, according to the example embodiment illustrated in FIG. **2**.

FIG. **5** illustrates another elevated operational side view of the drop-in system for a firearms trigger, according to the example embodiment illustrated in FIG. **2**.

FIG. **6** illustrates a bottom view of the drop-in system for a firearms trigger, according to the example embodiment illustrated in FIG. **2**.

The drawing illustrates a process for making the invention according to an example embodiment of the present inventive concept and is not to be considered limiting in its scope, as the overall inventive concept may admit to other equally effective embodiments.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

Reference will now be made in detail to the embodiments of the present general inventive concept, examples of which are illustrated in the accompanying drawings, wherein like reference numerals refer to the like elements throughout. The embodiments are described below in order to explain the present general inventive concept while referring to the figures. Also, while describing the present general inventive concept, detailed descriptions about related well-known functions or configurations that may diminish the clarity of the points of the present general inventive concept are omitted.

It will be understood that although the terms “first” and “second” are used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another element. Thus, a first element could be termed a second element, and similarly, a second element may be termed a first element without departing from the teachings of this disclosure.

Expressions such as “at least one of,” when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list.

All terms including descriptive or technical terms which are used herein should be construed as having meanings that are obvious to one of ordinary skill in the art. However, the terms may have different meanings according to the intention of the lexicographer, case precedents, or the appearance of new technologies. Also, some terms may be arbitrarily selected by the inventors, and in this case, the meaning of the selected terms will be described in detail in the detailed description herein. Thus, the terms used herein should be defined based on the generally defined meaning of the terms together with the description throughout this specification.

Hereinafter, one or more exemplary embodiments of the present general inventive concept will be described in detail with reference to accompanying drawings.

Example embodiments of the present general inventive concept are directed to a drop-in system for a firearms trigger which improves the resetting of the trigger upon firing a cartridge in a chamber of the firearm.

FIG. **2** illustrates an exploded perspective view of a drop-in system **100** for enhancement of a firearms trigger, according to an example embodiment of the present inventive concept. As illustrated in FIG. **2**, the drop-in system **100** for enhancement of a firearms trigger according to this example embodiment can include a trigger-locking block **102**, a hump block **104**, a trigger-locking block transfer bar **106** and an axis retainer plate **108**. Depending on the size of a trigger axis pin **124*cl*** (see FIG. **3**) that holds the trigger in a rotational position, the drop-in system **100** can also include a shortened adaptor sleeve **104*c*** to assist in a secure fitting of the hump block **104** onto the trigger axis pin **124*cl*** via a hump block hole **104*a*** extending therethrough. Positioning of the drop-in system **100** and each of the parts thereof into a receiver of a semiautomatic firearm are described below with respect to FIGS. **3-6**, as well as operations of each of the parts of the drop-in system **100** within a semiautomatic firearm receiver **118**.

FIG. **3** illustrates an operational plan view of the drop-in system **100** for a semiautomatic firearm, according to the example embodiment illustrated in FIG. **2**. The standard trigger **124** for a semiautomatic firearm generally includes a rear section **124*a***, a cradle section **124*b***, a hole **124*c*** extending through the sides of the cradle section **124*b***, a trigger axis pin **124*cl*** to hold the trigger **124** in place while being pivotal about the trigger pin **124*c*1**, and a hammer engagement hook **124*d*** (see also FIG. **4**) disposed within a semiautomatic firearms receiver **118**. The hump block **104** according to the present example embodiment can be placed into the cradle **124*b*** of the trigger **124** and can include a bottom portion **104*b*** and a top portion **104*c***, as well as the hole **104*a*** to receive the trigger axis pin **124*cl*** therethrough. Each of the sides of the cradle section **124*b*** include a trigger pin reception hole **124*c*** to receive the trigger axis pin **124*cl*** therethrough to enable pivotal movement back and forth of the trigger **124** while keeping the trigger **124** in place. The trigger axis pin **124*cl*** can also be inserted through the hole **104*a*** (see FIG. **2**) formed through the hump block **104** to retain the hump block **104** within the trigger cradle **124*b***, and also to enable the hump block **104** to be pivotal about the trigger axis pin **124*c*1**. In this configuration the hump block **104** can rock back and forth (i.e., pivot) about the trigger axis pin **124*c*1**. In the case where the diameter of the trigger axis pin **124*cl*** is smaller than the diameter of the hole extending through the hump block **104** a shortened pin adaptor sleeve **104*c*** can be inserted within the hole **104*a*** through the hump block **104**. The pin adaptor sleeve **104*c*** is configured to tightly fit within the hole **104*a*** formed through the hump block **104** and to have an inner diameter slightly larger than the diameter of the trigger axis pin **124*cl*** to enable the hump block **104** to freely rock back and forth (i.e., pivot) on the trigger axis pin **124*c*1**.

The trigger-locking block **102** is configured to be pivotally fixed in place by a well-known firearm safety device **126**, which is included with all or most semiautomatic firearms in order to disable and enable a user to fire the semiautomatic firearm by blocking and unblocking the user's ability to pull the trigger **124**. More specifically, the firearm safety device **126** generally includes a safety device axis pin **126*a*** that extends from a side thereof that is inserted through a first hole extending through one side of the semiautomatic firearms receiver **118**, through a hole extending through a retainer plate (not illustrated), and then snaps into a second hole extending through a second opposite side of the semiautomatic firearms receiver **118**. The safety device **126** is commonly provided as a pivot switch that will pivot the safety device **126** back and forth to either be positioned away from the rear end **124*a*** of the trigger **124** or directly over the rear end **124*a*** of the trigger **124**.

According to an example embodiment, the standard retainer plate can be first replaced with the specifically configured axis retainer plate **108** according to an example embodiment of the present inventive concept, as illustrated in FIG. **2**. The axis retainer plate **108** can be used to replace the standard retainer plate by removing the standard retainer plate and placing the axis retainer plate **108** within the semiautomatic firearm receiver **118**. More specifically the axis retainer plate **108** is configured to have a notch **108*a*** formed into a first end thereof such that the notch **108*a*** is inserted over a hammer axis pin **122*a*** in which a hammer **122** of a semiautomatic firearm rotates about. The axis retainer plate **108** also includes a hole **108*b*** extending through an opposite end thereof. The end of the safety device axis pin **126*a*** (opposite the side in which the safety device is connected) can be extended through the hole **108*b*** in the axis retainer plate **108** after the safety device axis pin **126*a*** is inserted through the first hole in the semiautomatic receiver and through a hole **102*a*** extending through the trigger-locking block **102**. The end of the safety device axis pin **126*a*** can then be snapped into the second hole extending through a second opposite side of the semiautomatic firearms receiver **118**. The axis retainer plate **108** also includes a bend **108*c*** disposed at approximately a middle section thereof to provide a spring action, as described in more detail below.

FIG. **4** illustrates an elevated operational side view of the drop-in system **100** for a firearms trigger, according to the example embodiment illustrated in FIG. **2**. FIG. **5** illustrates another elevated operational side view of the drop-in system for a firearms trigger, according to the example embodiment illustrated in FIG. **2**.

Referring to FIGS. **2-4**, a second end of the axis retainer plate **108** is configured to have a hole **108*b*** extending therethrough to fixedly receive the safety device axis pin **126*a*** which extends from a side of the firearm safety **126**. The trigger-locking block **102** is configured to have a hole **102*a*** extending therethrough in which the safety device axis pin **126*a*** will extend through such that the trigger-locking block **102** will rotate about the safety device axis pin **126*a*** of the firearm safety **126**. The trigger-locking block **102** can be configured to include an upper notch **102*b*** and a lower notch **102*c***, which is described in more detail below. The trigger-locking block **102** is also configured to include a trigger-locking block limit protrusion (or cylinder) **102*d*** (see FIG. **5**), which is also described in more detail below with reference to FIG. **5**. The trigger-locking block **102** can rotate back and forth on the safety device axis pin **126*a*** within a certain range of rotational motion due to the trigger-locking block limit protrusion **102*d***, as well as friction applied to the side of the trigger-locking block **102** by the bend **108*c*** formed in the axis retainer plate **108**, as is described in more detail below.

Generally, a semiautomatic firearm includes a bolt carrier **120** and a bolt **120*a***. Along the internal side of the semiautomatic firearm there is a bolt carrier guide **120*b***. The trigger-locking block transfer bar **106** is configured to be inserted into the semiautomatic firearm such that the trigger-locking block transfer bar **106** rests on and slides along the bolt carrier guide **120*b*** (see FIG. **5**). The trigger-locking block transfer bar **106** is configured to have a folded portion **106*a*** at a first end thereof and a tongue **106*b*** extending away from a side thereof close to a second end of the trigger-locking block transfer bar **106**.

Once the trigger-locking block **102**, the hump block **104**, the trigger-locking block transfer bar **106** and the axis retainer plate **108** are placed within the receiver of the semiautomatic firearm the process of firing and reloading the semiautomatic firearm acts as follows. When the semiautomatic firearm is placed in a cocked position a hammer lip **122*b*** (see FIG. **4**) will become engaged by the hammer engagement hook **124*d*** of the trigger **124**, thus preventing the hammer **122** from springing forward and hitting a firing pin (not illustrated for brevity of the detailed description). The firing pin hits the back of a round (cartridge) disposed in the chamber **128** and fires the round when the hammer **122** strikes the firing pin. When the trigger **124** is pulled the hammer **122** will become released from the hammer engagement hook **124*d*** and spring forward to strike the firing pin. This action will begin the cycling process described below.

The cycling process of a semiautomatic firearm with the drop-in system **100** for a semiautomatic firearms trigger, according to the present inventive concept, begins with the bolt carrier **120** traveling rearward as a result of the force of gas traveling through a gas tube to a piston at the front of the semiautomatic firearm. The gas traveling through the gas tube will reach the piston, which in turn will move the bolt carrier **120** backwards. As the bolt carrier travels backwards it will collide with the hammer **122**. The hammer **122** will then be forced to pivot rearward and collide with the top portion **104**c of the hump block **104**. As described above, the hump block **104** is pivotally fixed within the trigger cradle **124**b on the trigger axis pin **124**cl. Also as pointed out above, the hump block **104** can either be pivotally fixed directly on the trigger axis pin **124**cl or pivotally fixed on the trigger axis pin **124**cl with the aid of the axis adaptor sleeve **104**c, which is designed to take up space within the hole **104**a of the hump block **104**. The trigger assembly **124** will also be forced to pivot rearward by the hammer **122** colliding against the hump block **104**, since the hump block **104** will pivot back and the bottom section **104**b of the hump block **104** will push the rear end **124**a of the trigger down. At this point a hammer lip **122**b extending from the hammer **122** will become engaged by the hammer engagement hook **124**d on the trigger **124**. The hammer engagement hook **124**d will engage with the hammer lip **122**b and hold the hammer **122** from springing forward. The bolt carrier **120** will continue to travel rearward and the bolt **120**a on the bolt carrier **120** will collide with the upper notch **102**b of the trigger-locking block **102**. The trigger-locking block **102** will then pivot rearward on the safety axis pin **126**a, which extends between the firearm safety **126** and the axis retainer plate **108**. The trigger-locking block **102** will pivot rearward with friction applied to one side of the trigger-locking block **102** from the inward bend **108**c of the axis retainer plate **108** making contact with the trigger-locking block **102** until the trigger-locking block limit protrusion **102**d hits a bottom of the axis retainer plate **108** at a location where the bend **108**c extends inward (see FIG. **5**). Thus, the trigger-locking block **102** will pivot counterclockwise (with reference to FIGS. **3** and **4**) until the lower notch **102**c of the trigger-locking block **102** collides with the rear end **124**a of the trigger **124** and will push the rear end **124**a of the trigger **124** down to lock the trigger **124** in place. As pointed out above, the axis retainer plate **108** will limit the pivotal movement of the trigger-locking block **102** as a result of the trigger-locking block limit protrusion **102**d hitting the bottom of the axis retainer plate **108** at a location where the bend **108**c extends inward to catch the trigger-locking block limit protrusion **102**d (see FIG. **5**).

Once the bolt carrier **120** completes its travel rearward it will begin to travel in the forward direction (the "front" direction of the semiautomatic firearm) as a result of a spring action force applied by a well-known coil spring (not illustrated to provide brevity to the detailed description). As the bolt carrier **120** continues to travel forward a shoulder of the carrier bolt **120** will collide with the folded end **106**a of the trigger-locking block transfer bar **106**. The trigger-locking block transfer bar **106** will then be forced to slide forward along the carrier bolt guide **120**b formed along an inner side of the semiautomatic firearm receiver **118**. As a result of the forward sliding movement of the trigger-locking block transfer bar **106** the tongue **106**b of the trigger-locking block transfer bar **106** will collide with the upper notch **102**b of the trigger-locking block **102** and cause the trigger-locking block **102** to pivot clockwise (with reference to FIGS. **3** and **4**), thus unlocking the trigger **124** (the lower notch **102**c of the trigger-locking block **102** will rotate away from the rear end **124**a of the trigger **124**) from the trigger-locking block **102**. At this point the trigger **124** will be released from the trigger-locking block **102** and become enabled to be pulled again to restart the firing cycle.

FIG. **6** illustrates a bottom view of the drop-in system for a firearms trigger, according to the example embodiment illustrated in FIG. **2**. As illustrated from a bottom view, the trigger-locking block transfer bar **106** slides along an inner side of the receiver **118** and the axis retainer plate **108** retains the trigger-locking block **102** in place with the rotational assistance of the safety axis pin **126**a, which extends from the firearm safety **126**.

Although a few embodiments of the present general inventive concept have been shown and described, it will be appreciated by those skilled in the art that changes may be made in these embodiments without departing from the principles and spirit of the general inventive concept, the scope of which is defined in the appended claims and their equivalents.

The invention claimed is:

1. A drop-in system for a semiautomatic firearm that includes a trigger assembly pivotal about a trigger axis pin, a bolt carrier that slides back and forth along a receiver of the semiautomatic firearm, a spring-loaded hammer that rotates about a hammer axis pin, a firearm safety including a corresponding safety axis pin extending therefrom, and a bolt carrier guide that extends along an interior of the receiver to guide the bolt carrier back and forth, the drop-in system comprising:

an elongated axis retainer plate having one end configured to be fixed to the hammer axis pin and a second end configured to be fixed to the safety axis pin;

a trigger-locking block configured to pivot about the firearm safety axis pin, the trigger-locking block including a limit protrusion extending from a lower end thereof to contact the elongated axis retainer plate to limit rearward pivotal movement of the trigger-locking block;

a hump block configured to pivot about the trigger pin independently of pivotal movement of the trigger assembly; and

a trigger-locking block transfer bar configured to slide along the bolt carrier guide and including a limit protrusion extending from a rear bottom portion thereof.

2. The drop-in system for a semiautomatic firearm according to claim **1**, wherein when the bolt carrier travels in a rearward direction along the receiver due to firing of the semiautomatic firearm:

the bolt carrier collides with the hammer;

the hammer rotates rearward and collides with the hump block;

the hump block pivots rearward and collides with the trigger assembly to rotate the trigger assembly rearward; and

the bolt carrier collides with the trigger-locking block to rotate the trigger-locking block rearward to lock a rear end of the trigger assembly.

3. The drop-in system for a semiautomatic firearm according to claim **2**, wherein after the bolt carrier completes the travel rearward the bolt carrier springs in a forward direction to collide with the limit protrusion of the trigger-locking block transfer bar to slide the trigger-locking bar in a forward direction such that the limit protrusion collides with a top portion of the trigger-locking block to pivot the trigger-locking block in a forward direction to release the rear end of the trigger assembly from the locked position.

4. The drop-in system for a semiautomatic firearm according to claim **3**, wherein the elongated axis retainer plate comprises:

a notch formed at a first end thereof to connect with the hammer axis pin;

a hole extending through a second opposite end thereof to receive the safety axis pin therethrough; and a bend formed at a middle portion thereof to extend inward and apply a friction to a side of the trigger-locking block and to stop the limit protrusion to limit the rearward rotation of the trigger-locking block.

5. The drop-in system for a semiautomatic firearm according to claim **4**, wherein the elongated axis retainer plate is disposed in the receiver by the notch formed at the first end thereof being fixed over the hammer axis pin and the safety axis pin being slid through a first hole extending through a first side of the receiver, through a hole extending through the trigger-locking block, through the hole extending through the second end of the elongated axis retainer plate, and an end of the safety axis pin being snapped into a second hole extending through a second side of the receiver.

6. A drop-in system for a semiautomatic firearm configured to be placed within a receiver of the semiautomatic firearm, comprising: an axis retainer plate configured to be disposed between a hammer axis pin and safety axis pin; a hump block configured to pivot about a trigger axis pin of a trigger to pivot the trigger backward; a trigger-locking block configured to pivot about the safety axis pin in a first direction until making contact with the elongated axis retainer plate and locking the trigger from being pulled; and a trigger-locking block transfer bar configured to slide along an inner wall of the receiver and including a limit protrusion extending from a rear bottom portion thereof to pivot the trigger-locking block in a second direction to unlock the trigger.

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